Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/17/2005
PI Name:	Puri, Ishwar K. Ph.D.		
Project Title:	The Extinction of Low Strain Rate Diffusion Flames in an Agent in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	esmhead@vt.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	540-231-3243
Organization Name:	Virginia Polytechnic Institute and State University		
PI Address 1:	Department of Engineering Science and Mechanics		
PI Address 2:	223 Norris Hall - MC 0219		
PI Web Page:	http://www.esm.vt.edu/		
City:	Blacksburg	**State:**	VA
Zip Code:	24061	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GB64G		
Performance Goal No.:			
Performance Goal Text:			

This proposal is a request for continued funding for an experimental and computational study that has been supported through the NASA microgravity grants program from February 1999 through February 2003. The work has investigated the structure and suppression of low strain rate non-premixed flames under normal gravity and microgravity conditions. The topic is relevant from both a fire safety and a fundamental perspective. Unwanted fires typically occur due to non-premixed combustion. In terms of fire protection, the agent concentration required to achieve suppression of non-premixed flames is important.
In the current study, a drop tower rig and associated diagnostics have been prepared to conduct suppression experiments in the 2.2 s NASA-Glenn drop tower. Preliminary experiments have measured the critical N2 concentration required for flame suppression when N2 is added to the fuel stream of methane-air diffusion flames. Normal-gravity measurements have been conducted for comparison with the microgravity results. In addition, the measurements have been compared

	to computational results. Measurements were compared to a previously developed one-dimensional flamelet code that was modified to include radiative heat loss by participating gas species. A transient axisymmetric flame simulation is being developed. The preliminary version of the code uses a mixture fraction approach to simulate the dynamics of flame transition from normal to microgravity in drop tower experiments.
Task Description:	The work proposed here is an extension of the current study. The proposed research addresses several areas beyond what is expected to be completed by the end of the current grant period. The proposed study includes: (1) broadening the scope of the tests to include a variety of fuels and agents. Fuels other than CH4 that are more representative of actual fires (higher sooting tendency) and agents other than N2 that are representative of suppressants in use on space platforms (CO2, CF3Br, etc) will be investigated. It is expected that a database of suppression requirements for a matrix of fuels and suppressants will be useful to NASA for sizing agent mass requirements for total flooding applications; (2) diagnostic capabilities will be added to the microgravity experimental rig including (a) the use of radiometers to measure radiative emission for comparison to the simulations and (b) the use of two-color pyrometry to measure soot volume fractions; (3) the flame simulations will be enhanced such that (a) the one-dimensional flamelet code will be modified to include soot and the radiative emission due to particulates when appropriate. The computed flame structure will be validated by comparison with the measured distributions of temperature, heat flux, and soot; and (b) The two-dimensional transient flame simulation will be improved by inclusion of finite rate kinetics and radiative flame emission. The proposed model would then allow insight into the dynamics of flame extinction. The objective is to characterize the influence of thermal radiation on flame extinction. The task will enable us to beteer engineer fire safety.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fire safety is of concern in both space and earth enviironments.
Task Progress:	We have characterized the influence of thermal radiation with various diluents on flame stability. How a thermally radiating diluent is introduced is important. For instance, we find that when carbon dioxide is injected directly into the fuel-rich central jet of a coflow flame, it is more effective at suppressing the flame than when it is introduced into the air coflow. This influence changes in normal and microgravity.
Bibliography Type:	Description: (Last Updated: 12/29/2005)
Articles in Peer-reviewed Journals	ANDREW LOCK1, ISHWAR K. PURI2, SURESH K. AGGARWAL1 and UDAY HEGDE3 1Mechanical Engineering, University of Illinois at Chicago, Chicago IL 2Engineering Science and Mechanics, Virginia Polytechnic Institute, Blacksburg, VA 3National Center for Microgravity Research, NASA Glenn Research Center, Cleveland OH "Flame Liftoff and Blowout of Partially Premixed Flames in Normal Gravity and Microgravity" Proceedings of the International Symposium on Fire Safety Science, Vol. 8 , Sep-2005